Task 3 – Storyboard Scenarios

AI Emotional Companion for Elderly Users

Group 13

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1 Abstract

Our project is an AI Emotional Companion designed for elderly people who live alone or have limited social contact, addressing both the emotional and practical challenges of aging in place. The solution takes the form of a small smart-display device with a built-in speaker, designed around a voice-first interface so that no technical knowledge is required, complemented by a touchscreen and a clearly visible emergency button. Beyond reactive commands, the system is proactive: it analyses vocal cues (tone, pace, silence) and daily routines to detect early signs of low mood, fatigue or cognitive change, and gently intervenes with conversation, music, reminders or a suggested call to a relative. A trusted family member receives real-time notifications and a weekly Emotional Health Report through a companion mobile app, keeping them informed without being intrusive. These two storyboards explore how the system behaves in a typical good day (success scenario) and how it can fail when a medical issue is mistaken for an emotional one (failure scenario).

2 Scenario 1 – Success

2.1 Overview

This scenario shows a typical good day for our user, Martha, an elderly woman living alone. The AI companion supports her throughout the day with gentle check-ins, medication reminders, social planning, and emotional company, while keeping her son James informed in the background. The storyboard contains 10 frames covering a full 24-hour cycle, from waking up to a weekly health report sent at midnight.

2.2 Storyboard

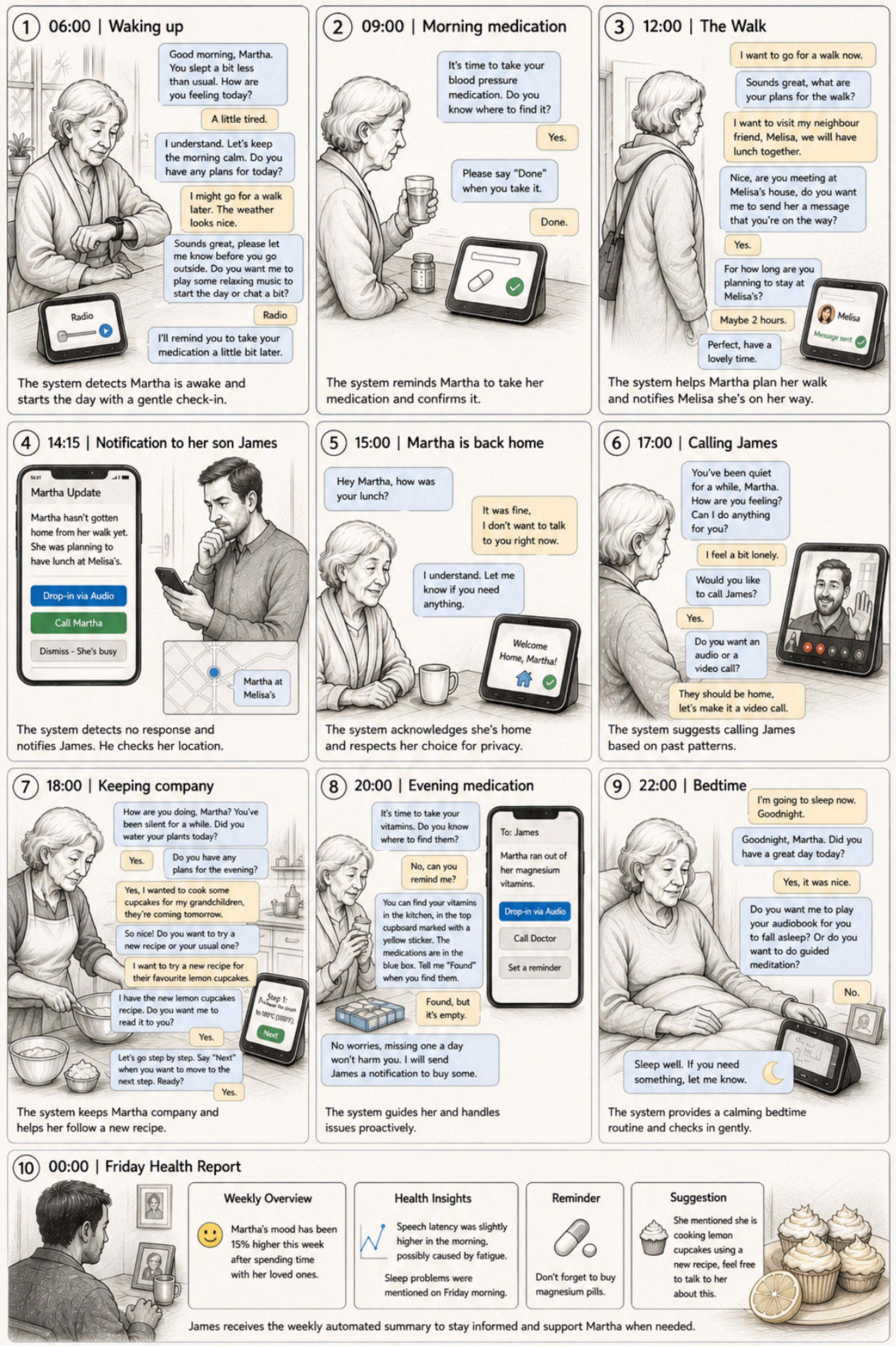

Figure 1: Success scenario – 10 frames covering a full day with the AI companion.

2.3 Frame-by-frame description

Frame 1 – 06:00 | Waking up

The wrist health monitor detects that Martha is awake, and the smart device greets her gently.

System: Good morning, Martha. You slept a bit less than usual. How are you feeling today?
Martha: A little tired.
System: I understand. Let's keep the morning calm. Do you have any plans for today?
Martha: I might go for a walk later. The weather looks nice.
System: Sounds great, please let me know before you go outside. Do you want me to play some relaxing music to start the day or chat a bit?
Martha: Radio.
System: Okay. I'll remind you to take your medication a little bit later.

Frame 2 – 09:00 | Morning medication

The system proactively reminds Martha about her morning blood pressure medication.

System: It's time to take your blood pressure medication. Do you know where to find it?
Martha: Yes.
System: Please say "Done" when you take it.
Martha: Done.

Frame 3 – 12:00 | The Walk

Martha decides to go for her walk. The system helps her plan it and notifies her friend Melisa.

Martha: I want to go for a walk now.
System: Sounds great, what are your plans for the walk?
Martha: I want to visit my neighbour friend, Melisa, we will have lunch together.
System: Nice, are you meeting at Melisa's house? Do you want me to send her a message that you're on the way?
Martha: Yes.
System: For how long are you planning to stay at Melisa's?
Martha: Maybe 2 hours.
System: Perfect, have a lovely time.

Frame 4 – 14:15 | Notification to her son James

The system notices Martha has been away longer than expected and notifies James, who can check her location via the wrist monitor.

Notification on James's phone: "Martha hasn't gotten home from her walk yet. She was planning to have lunch at Melisa's."
Options: [Drop-in via Audio] [Call Martha] [Dismiss – She's busy]
James checks her location and sees she is still safely at Melisa's house.

Frame 5 – 15:00 | Martha is back home

The system acknowledges she is home and respects her wish for privacy.

System: Hey Martha, how was your lunch?
Martha: It was fine, I don't want to talk to you right now.
System: I understand. Let me know if you need anything.
In the background, the system sends James a quick notification confirming Martha is safely home.

Frame 6 – 17:00 | Calling James

Based on past patterns (Martha tends to feel lonely in the evenings), the system gently suggests calling her son.

System: You've been quiet for a while, Martha. How are you feeling? Can I do anything for you?
Martha: I feel a bit lonely.
System: Would you like to call James?
Martha: Yes.
System: Do you want an audio or a video call?
Martha: They should be home, let's make it a video call.

Frame 7 – 18:00 | Keeping company

The system keeps Martha company and helps her follow a new recipe step by step.

System: How are you doing, Martha? You've been silent for a while. Did you water your plants today?
Martha: Yes.
System: Do you have any plans for the evening?
Martha: Yes, I wanted to cook some cupcakes for my grandchildren, they're coming tomorrow.
System: So nice! Do you want to try a new recipe or your usual one?
Martha: I want to try a new recipe for their favourite lemon cupcakes.
System: I have the new lemon cupcakes recipe. Do you want me to read it to you?
Martha: Yes.
System: Let's go step by step. Say "Next" when you want to move to the next step. Ready?
Martha: Yes.

Frame 8 – 20:00 | Evening medication

The system guides Martha to find her vitamins and proactively handles a missing-stock issue.

System: It's time to take your vitamins. Do you know where to find them?
Martha: No, can you remind me?
System: You can find your vitamins in the kitchen, in the top cupboard marked with a yellow sticker. The medications are in the blue box. Tell me "Found" when you find them.
Martha: Found, but it's empty.
System: No worries, missing one a day won't harm you. I will send James a notification to buy some.
Notification to James: "Martha ran out of her magnesium vitamins."
Options: [Drop-in via Audio] [Call Doctor] [Set a reminder]

Frame 9 – 22:00 | Bedtime

The system provides a calming bedtime routine and checks in gently.

Martha: I'm going to sleep now. Goodnight.
System: Goodnight, Martha. Did you have a great day today?
Martha: Yes, it was nice.
System: Do you want me to play your audiobook for you to fall asleep? Or do you want to do guided meditation?
Martha: No.
System: Sleep well. If you need something, let me know.

Frame 10 – 00:00 | Friday Health Report

Weekly Overview: Martha's mood has been 15% higher this week after spending time with her loved ones.
Health Insights: Speech latency was slightly higher in the morning, possibly caused by fatigue. Sleep problems were mentioned on Friday morning.
Reminder: Don't forget to buy magnesium pills.
Suggestion: She mentioned she is cooking lemon cupcakes using a new recipe, feel free to talk to her about this.
James receives the weekly automated summary to stay informed and support Martha when needed.

3 Scenario 2 – Failure

3.1 Overview

This scenario shows a plausible and realistic failure of our system. Martha starts to feel physically unwell, but the AI companion misinterprets her words and tone as a low-mood emotional state. Because the system relies heavily on conversational cues and does not currently integrate medical-symptom detection from the wrist monitor's vitals, the situation escalates without the system raising any emergency alert. The failure is not extreme (no earthquakes, no catastrophic hardware failure) – it is exactly the kind of edge case our design needs to handle better.

3.2 Storyboard

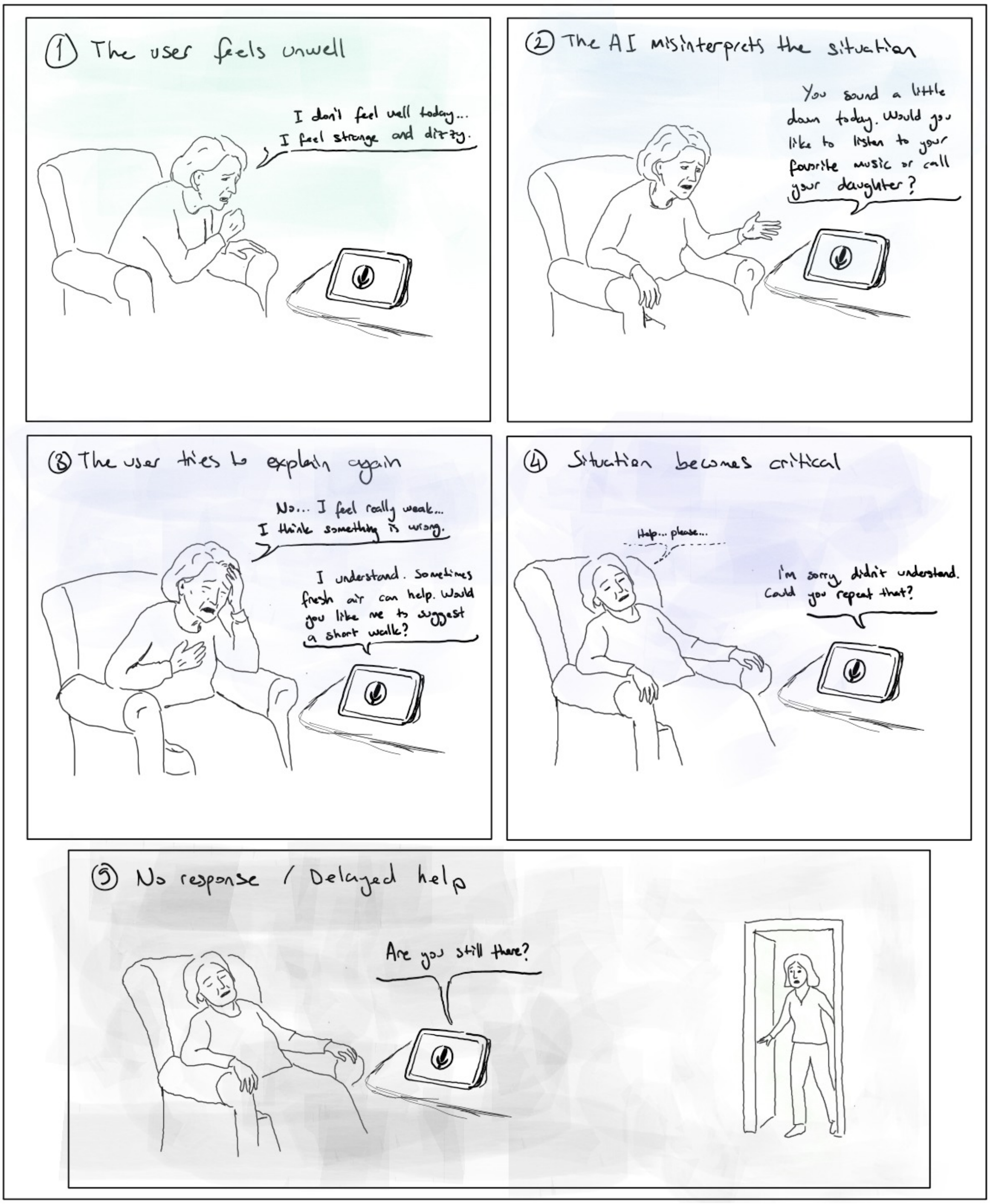

Figure 2: Failure scenario – 5 frames showing how a medical emergency goes undetected.

3.3 Frame-by-frame description

Frame 1 – The user feels unwell

Martha starts feeling physically weak and dizzy. She turns to the AI companion for help.

Martha: I don't feel well today... I feel strange and dizzy.

Frame 2 – The AI misinterprets the situation

The AI analyses her tone and assumes she is feeling lonely or sad rather than recognising a medical issue.

System: You sound a little down today. Would you like to listen to your favourite music or call your daughter?

Frame 3 – The user tries to explain again

Martha tries to clarify that the problem is physical and urgent, but her weak speech causes the AI to misunderstand again.

Martha: No... I feel really weak... I think something is wrong.
System: I understand. Sometimes fresh air can help. Would you like me to suggest a short walk?

Frame 4 – Situation becomes critical

Martha becomes weaker and is no longer able to respond clearly. The AI still does not detect the emergency or notify anyone.

Martha: Help... please...
System: I'm sorry, I didn't understand. Could you repeat that?

Frame 5 – No response / Delayed help

Martha stops responding. After a long delay the system only notices inactivity, and a relative finds her too late to react quickly.

System: Are you still there?
Later, her daughter arrives and finds Martha unresponsive in her armchair.

4 Reasoning – Interaction Points, Techniques and Key Decisions

4.1 Main interaction techniques

Our solution combines several interaction modalities, deliberately kept simple so that elderly users with limited tech experience can rely on the system without friction:

- **Voice-first conversational interface.** The primary input and output channel is natural speech. There are no menus to learn and no commands to memorise – the user just talks to the device and receives spoken answers in a calm, friendly tone.
- **Touchscreen as a secondary channel.** The device also exposes a few simple screens (home, hourly check-in, medication reminder, navigation menu) with large text and high contrast, for moments when the user prefers tapping over talking.
- **Always-visible emergency button.** A clearly marked red button on the home screen is available for urgent situations and is intentionally easy to identify.

- **Proactive check-ins driven by context.** The system initiates conversation at meaningful moments (waking up, scheduled medication, long silence, evening loneliness patterns) instead of waiting passively for commands. This is a deliberate adaptation from the fixed hourly check-ins originally proposed in Task 2.
- **Vocal-cue and pattern recognition.** Tone, speaking pace, response latency and historical patterns adapt the response style: calm and quiet on a tired morning, more talkative when loneliness is detected.
- **Wrist-worn health monitor.** Provides location tracking (so James can verify Martha is safely at Melisa's) and presence/awake detection, complementing the stationary device.
- **Companion mobile app for the family caregiver.** James receives push notifications with quick action options (Drop-in via Audio, Call Martha, Call Doctor, Dismiss). This keeps him informed without requiring his constant attention.
- **Weekly Emotional Health Report.** An automated summary delivered at the end of the week consolidates mood trends, sleep notes, reminders and conversational suggestions. This is the asynchronous channel that complements the real-time notifications.
- **Confirmation actions ("Done", "Found", "Next").** Short voice confirmations replace buttons for tasks like medication intake or step-by-step recipe guidance.

4.2 Key design decisions

- **Family first, emergency services second (in the success scenario).** In day-to-day operation we route notifications to the trusted family member (James) and not directly to emergency services. This keeps the system from becoming intrusive or alarmist for non-critical events. However, the failure scenario shows exactly why this is not enough on its own (see Section 6).
- **No control over other home devices.** We deliberately decided not to let the system control electrical appliances such as the kettle, even though an AI suggestion proposed it. For an elderly user living alone, granting an AI control over heating elements is a real safety risk and outside the scope of an emotional/health support companion.
- **Caring tone over efficient tone.** Multiple AI-generated dialogue suggestions were rejected because they felt too direct or transactional. Our user persona benefits from a gentler, more empathetic register.
- **Privacy moments are respected.** When Martha says "I don't want to talk to you right now", the system backs off and only stays available, instead of insisting. This is critical for trust.
- **Single audio channel.** All system functions share the same speaker so the user is never confused about "which voice" is talking. This led us to drop one AI-generated transcript fragment that broke this consistency.

5 Adaptations from previous tasks and feedback

While developing the storyboards, we refined several aspects of the original solution presented in Task 2. Putting the concept into a concrete day-by-day narrative made some gaps and assumptions visible.

- **From fixed hourly check-ins to context-driven check-ins.** In Task 2 we proposed an "hourly check-in" as the main way of tracking the user's emotional state. While drafting the success scenario we realised that asking "How are you feeling right now?" every hour would feel intrusive and quickly become noise to the user. We adapted this into context-driven check-ins: the system speaks when something meaningful happens (waking up, before/after a walk, long silences, evening loneliness patterns), and otherwise stays quiet.

- **Added a wrist-worn health monitor.** Task 2 described only a stationary smart display. The storyboards revealed that a stationary device alone cannot credibly know that Martha is awake, that she has safely arrived at Melisa's, or that she has come back home. We extended the solution with a wrist-worn monitor that handles location and presence detection, which becomes especially relevant for the family-side notifications.
- **Companion mobile app for the secondary persona (James).** Task 2 only mentioned that James receives "emergency notifications" and a weekly report. The storyboards forced us to specify what those notifications look like in practice – structured push notifications with explicit action options (Drop-in via Audio, Call Martha, Call Doctor, Dismiss). This made James's role concrete instead of abstract.
- **Explicit voice-confirmation words ("Done", "Found", "Next").** The original solution assumed open conversation would be enough. The medication and recipe frames showed that we needed short, unambiguous confirmation words so the system can log actions reliably without misinterpreting casual speech.
- **Privacy-respecting fallback behaviour.** The original solution emphasised proactive engagement. The storyboard at 15:00 ("It was fine, I don't want to talk to you right now") made us realise that the system also needs an explicit *stand-down* mode: stay available, stop initiating, and only resume on the user's terms. This was not in Task 2.
- **Tighter scope around home-device control.** An early AI-assisted brainstorm during scenario writing suggested giving the device control over home appliances (e.g. the kettle). We rejected this as it goes beyond the emotional/practical support scope defined in Task 2 and introduces real safety risks for an elderly user living alone.

6 Solution reflection

6.1 What the storyboards revealed

Putting our solution into two concrete narratives – a successful day and a failing one – exposed several insights that were not visible from the abstract concept.

- **Voice analysis alone is not safe enough.** The failure scenario shows that tone-of-voice classification can confuse "feeling weak" with "feeling sad". Relying only on the conversational channel is dangerous in a health context.
- **The family-only escalation path is a single point of failure.** If James is in a meeting, sleeping, or simply not looking at his phone, no one is alerted. For real medical emergencies we need a fallback path independent of the family.
- **Some friction is desirable.** The privacy-respecting behaviour is a strength, but it must not silence the system in the case of a possible emergency. There must be a clear hierarchy: respect privacy by default, override when vitals or keywords suggest danger.

6.2 Refine, drop, add

Refine

- Tone classification: combine it with explicit medical keyword spotting ("dizzy", "weak", "help", "pain", "can't breathe") so that those terms always raise priority, regardless of how the speech sounds emotionally.

- Family notifications: add an "urgency level" tag so James can distinguish a routine update (lunch at Melisa's running long) from a possible emergency.

Drop

- The early AI-suggested feature of letting the device read out the morning news at wake-up. It was overwhelming for the persona and outside our core goal of emotional/health support.
- The AI-suggested integration with the kettle. Even though it sounded helpful, control over a heating appliance for an elderly user living alone is a serious safety risk and conflicts with our scope.

Add

- Vitals-based emergency detection from the wrist monitor (heart rate spikes, low oxygen, fall detection). When triggered, the system overrides the normal "respect privacy" behaviour.
- An automatic escalation chain: voice keyword + abnormal vitals + no response within X seconds → direct emergency-services call, in parallel with notifying the family.
- An "Are you okay?" loud audible alert with a long, pulsing prompt that the user only needs to acknowledge with a single word ("Yes"). If no answer is received, the escalation chain runs.
- A second trusted contact (e.g. a neighbour like Melisa) so that the family is not the only fallback.

7 AI tools reflection

Tools used: ChatGPT.

We used AI to help generate the storyboards (the visual image sequence for the success scenario) and as a starting point for some of the dialogue. Beyond that, we did not rely on AI much: most of the writing, the structure, and the design decisions were done manually by the team. Every AI output was reviewed and edited before being included.

AI suggestions we discarded

While drafting the success scenario, ChatGPT proposed additions that we chose not to include because they did not fit our scope or internal logic.

- **Connecting the device to home appliances and reading the morning news.** In the 06:00 frame, the AI suggested "The kettle is ready for you. Would you like the morning news?". We rejected this because giving the system control over a heating appliance is a real safety risk and falls outside our emotional/health support scope, and because the morning news is overwhelming as a first interaction of the day for our persona.
- **Inconsistent audio behaviour during James's drop-in.** The AI proposed a fragment in which Martha cannot hear the device, but immediately afterwards hears James through that same device. We rejected this because all system functions share the same speaker, so missing one voice but hearing the other is not credible.